PAYTON'S LiDAR SENSOR
LASER SPEED · TF03 LiDAR · DESIGNED BY PAYTON PITTS

Vault approach report
TF03 LiDAR · 1000 Hz · **±0.03 s · ±0.1 m/s**

POLE VAULT · RUNWAY · SHORT APPROACH

Demo Athlete

Nov 7, 2026 · 1:15 PM Tripod 100 ft from the back of the box

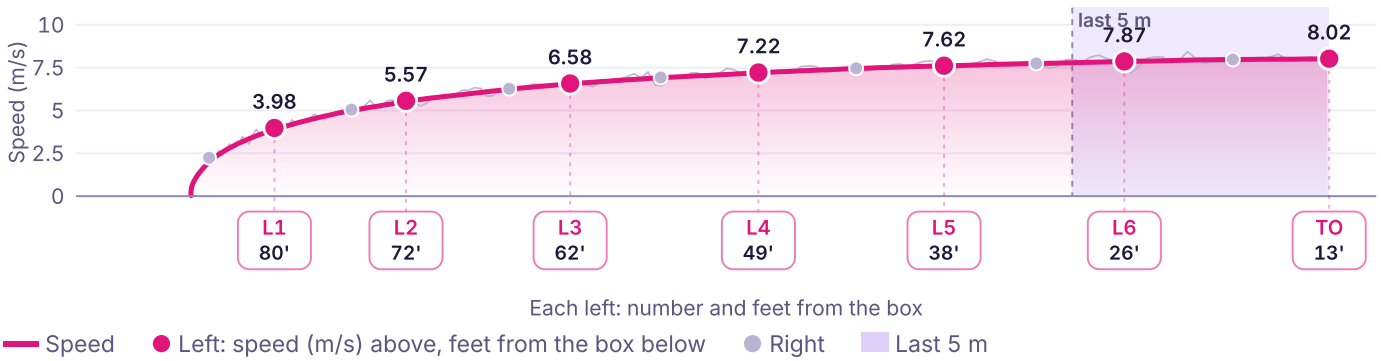
Last 5 m into takeoff	Top speed	Lefts	Takeoff ≈
7.94 m/s	**8.02 m/s**	**7**	**13' 0"**
5 m before: 7.60 m/s	91% of sprint top speed	14 steps	out from the box
+0.25 m/s speeding up into the box last ½ s before the plant	Start mark ≈ **86' 4"** from the box	Approach time **3.76 s** 72.6 ft run	Power **1020 W** 15.0 W/kg · peak horizontal

Summary · written by the app from this jump

- Demo's last 5 m into takeoff was **7.94 m/s**, built 0.34 into takeoff compared with the 5 m before it (7.60 m/s).
- Top speed on the runway was **8.02 m/s**, 91% of the best sprint top speed (8.81 m/s).
- Into the box: **+0.25 m/s**, speeding up into the box over the last half second before the plant.
- **7 lefts** (14 steps), counted back from the takeoff.

NEXT SESSION Still speeding up into the box: good. Keep the takeoff spot the same as speed rises, and check the mid mark stays put.

Speed into the takeoff



Step rhythm

#	FOOT	FROM BOX	STEP LENGTH	STEP TIME	SPEED (M/S)
1	R	84' 7"	–	–	2.24
2	L1	80' 5"	4' 2"	0.40 s	3.98
3	R	75' 6"	4' 11"	0.33 s	5.05
4	L2	72' 0"	3' 6"	0.20 s	5.57
5	R	65' 5"	6' 7"	0.34 s	6.27
6	L3	61' 6"	3' 11"	0.18 s	6.58
7	R	55' 9"	5' 9"	0.26 s	6.93
8	L4	49' 6"	6' 3"	0.27 s	7.22
9	R	43' 3"	6' 3"	0.26 s	7.46
10	L5	37' 8"	5' 7"	0.23 s	7.62
11	R	31' 9"	5' 11"	0.23 s	7.75
12	L6	26' 1"	5' 7"	0.22 s	7.87
13	R	19' 2"	6' 11"	0.27 s	7.98
14	L7 · TO	13' 0"	6' 2"	0.22 s	8.02

Speed you carry to the plant

Sprint top speed	8.81 m/s
Runway top speed	8.02 m/s
Last 5 m	7.94 m/s
Last 5 m as % of sprint top speed	90%
Speed at the last 3 lefts	7.62 → 7.87 → 8.02
Step length: last 6 vs 6 before	6' 1" vs 5' 2"
33 → 16 ft out from the box	7.92 m/s

Speed at each left

L1 · 80'	3.98
L2 · 72'	5.57
L3 · 62'	6.58
L4 · 49'	7.22
L5 · 38'	7.62
L6 · 26'	7.87
TO · 13'	8.02

Bars are measured against sprint top speed. They should keep growing to the takeoff.

How to use this as a vaulter

Last 5 m into takeoff
The speed you bring to the plant, and the speed the pole has to work with. Faster here, with the same takeoff, means a stiffer pole and a higher grip. This is the number to chase.

% of sprint top speed
How much of your max speed survives the pole carry. If it's low, the run-up is costing you speed: work on carry posture, relaxed arms and a steady rhythm. If it's high, the next jump in speed has to come from sprint training.

Speed at each left
Should keep rising, or at least hold, all the way to takeoff. A dip at the last 2–3 lefts means slowing down to find the box or reaching for the plant.

Step length (grey rights in between)
Steps should get a little shorter and quicker at the end, not longer. Overstriding in the last steps usually shows up as a speed drop into takeoff.

Lefts, takeoff and start mark
The same lefts with a takeoff that keeps moving means the run-up is changing. Use the takeoff distance to move the start mark, and keep the left count the same while speed improves.

Run-up length
The sprint report shows where you reach 98% of top speed. If your approach is shorter than that, you plant before you're at full speed. Add a left if you can still control it.